Syamdilya Nimmagadda

Email: nsyamdilya@gmail.com — **Mobile:** +1 (916) 666-3844 — **Location:** Sacramento, CA
LinkedIn: linkedin.com/in/syamdilya-nimmagadda/

Objective

Graduate in Computer Engineering with 2+ years hands-on experience in RTL design, static timing analysis (STA), hardware verification, and system architecture. Proficient in Verilog, SystemVerilog, and UVM for digital design and verification. Seeking full-time opportunities in semiconductor design and verification, focusing on delivering high-performance, innovative hardware solutions.

Education

Master of Science in Computer Engineering *Aug 2022 – Dec 2024*
California State University, Sacramento, CA – GPA: 3.48

B.Tech in Electronics and Communication Engineering *Jun 2018 – May 2022*
Shri Vishnu Engineering College for Women, India – GPA: 3.48

Experience

Design Verification Engineer *Feb 2026 – Present*
Viz Digital Marketing LLC

- Collaborated with design and verification teams to execute test cases, analyze simulation results, and support debugging activities for digital designs.
- Assisted in maintaining verification documentation and tracking issues to ensure compliance with design specifications.
- Supported regression testing and coverage analysis while contributing to timely project completion.

Design Verification Engineer *Feb 2025 – Jan 2026*
Pinnacle arc LLC, Denver, CO

- Verified Direct Memory Access (DMA) controller functionality including single transfer, burst, and scatter-gather modes.
- Developed UVM testbench components (agents, monitors, scoreboards) to verify DMA data transfers between peripherals and memory. Implemented constrained-random sequences to stress-test multiple DMA channels with back-to-back and overlapping transactions.
- Verified DMA protocol compliance for AHB interfaces including unaligned transfers, back-pressure handling, and error reporting.
- Created functional coverage (transfer size, burst type, channel priority) to ensure full corner-case coverage.
- Designed scoreboard and checkers for end-to-end data integrity across DMA transactions.
- Debugged DMA timing and arbitration issues at subsystem and SoC level in collaboration with design and integration teams. Achieved coverage closure greater than 95% on DMA verification by combining directed, random, and stress test scenarios.

Design Engineer Intern *Jun 2023 – Dec 2023*
Pinnacle arc LLC, Denver, CO

- Contributed to the Power Management Unit (PMU) Interface by developing control logic for clock gating, reset sequencing, and status monitoring, ensuring compliance with low-power design methodologies (UPF, CDC, RDC rules).
- Built Python automation scripts to compare register maps with architectural specifications, streamlining functional verification and reducing manual effort.
- Performed linting, synthesis preparation, and STA checks with Synopsys Design Compiler, analyzing timing and fixing violations to align with client's coding and quality guidelines.

Digital Design Engineer Intern *Apr 2021 – Sep 2021*
Cloud Software Solutions, Bengaluru, India

- Assisted the digital design team in validating and enhancing pre-developed Verilog modules such as ADC interfaces, simple filters, and PWM generators, ensuring functional correctness through simulation in Synopsys VCS.
- Wrote directed testbenches in Verilog to verify functional scenarios and corner cases and supported debugging of simulation mismatches alongside senior engineers.
- Created Python-based automation scripts to extract functional coverage and cross-check register specifications, reducing manual review effort and improving consistency.
- Contributed to timing and power analysis reviews of existing modules, applying clock gating and

reset domain handling concepts under guidance, while participating in lint and design rule checks to ensure reliability and compliance.

Technical Skills

Programming: C, C++.
Operating Systems: Linux.
Scripting Languages: Python, Perl, TCL.
HDL and Methodology : Verilog, SystemVerilog and UVM
Tools: Synopsys Design Compiler (VCS), Xilinx Vivado (NEXYS4 DDR), Quartus Prime (DE10-Lite), Questasim, SimVision, JasperGold, Jupyter Notebook, MATLAB.
Microsoft Tools: Visual Studio, Excel.
Communication Protocols: I2C, SPI, PCI, UART, AMBA.
Design and Verification: Finite State Machine (FSM) design, Digital Logic Design, Static Timing Analysis (STA), CPU Architecture, Design Rule Check, Logic Synthesis, Clock Domain Crossing.

Projects

Security-Based Implementation of Obfuscated 32-Bit ALU Architecture
- Architected the implementation of logic obfuscation in Verilog HDL to enhance security while maintaining full functionality in a 32-bit ALU design.
- Utilized the Vedic multiplication algorithm to minimize power consumption and design complexity while developing an optimized architecture for Binary-Coded Decimal (BCD) addition to enhance computational efficiency.

RTL Design and Verification of a 4-Port Switch/Router
- Co-implemented a parameterized 4-port switching datapath with per-port FIFOs and store-and-forward, coordinating pipeline adjustments based on STA checks.
- Built a UVM-style environment with constrained-random traffic, reporting coverage metrics and failures to speed debug cycles.
- Led verification efforts by integrating SVA assertions and Python-based log analysis to ensure consistent regression results.

SPI Slave with Flow Control
- Implemented an SPI Slave with protocol-level flow control using dual-clock RX/TX FIFOs and watermark-based backpressure; added a BUSY-token path for not-ready reads and status-first polling; performed basic STA/SDC and CDC hardening with synchronizers.
- Built a UVM-style testbench that randomly varies SPI modes and frame lengths and checks results with a self-checking scoreboard. Added SystemVerilog Assertions to enforce CS/SCK timing and prevent FIFO under/overflow, and used small Python scripts to plot throughput and FIFO usage.

PCIe Transaction Layer Endpoint
- Contributed to the design of a PCIe Endpoint handling Memory Read/Write requests, BAR0 address decoding, credit counters, and up to eight outstanding reads with proper completions.
- Implemented BAR0 decode logic, outstanding read tracking, and credit counter updates in Verilog HDL, and added targeted assertions to support verification.

Router 1x3 Design and Verification
- Designed and synthesized a 1x3 router in Verilog HDL with Register, FIFO, FSM, and Synchronizer submodules.Developed and validated models to ensure system accuracy and reliability through simulation in Xilinx Vivado and Synopsys VCS.
- Built a UVM-based verification environment, applied regression testing and SVA assertions, and achieved functional and code coverage closure for RTL sign-off.

Certifications

- **RTL-to-GDSII Flow v7.0** — Cadence.
- **Jasper Formal Fundamentals v2403** — Cadence.
- **SystemVerilog Assertions v5.1** — Cadence.

Additional Projects

- Third Eye for the Blind using Arduino (IoT and Embedded Systems)
- Predicting Energy Output of Wind Turbine using Weather Data (Machine Learning)
- Smart Medicine Reminder (IoT and Cloud Integration)